Participant A

Madelin Burt-D'Agnillo 00:44
How do you define open science?

00:58
Well, I guess instead of how I define, maybe I will answer how I understand open science because I don't think I can come up with anything concise enough to be a definition. I understand open science as openness through the whole process of creating research, from planning through to conducting data collections, through to analysis, through to sharing results and data and then having subsequent conversations about it. So openness at different stages, and obviously, it will be openness at different levels as well.

Madelin Burt-D'Agnillo 01:34
Excellent. That's a great understanding, I think, quasi definition. Okay. And what are your experiences with open science practices? How do you engage with them?

01:47
So when I intersect with open science, it's mostly at the end of the research process, when we talk about publishing, so most of my work has to do with open access publishing, I consider myself to be an open access advocate. [...] So where I intersect with open science is when the final product is already prepared when research has been conducted when research outputs have been put together in a manuscript, an article, a report, a thesis, and they are made open online.

Madelin Burt-D'Agnillo 02:49
Awesome, great. And question five is, where do you learn about open science practices?

02:57
Where do I learn about open science practices? It's been so long, I feel like I've always known about them. That's not true. So in terms of open access, which is my main focus, as I mentioned, I've learned about them when I first started this job. Before I started this job, I was also kind of working with repositories. So I learned about open access pretty much early on. And I have since been kind of developing my knowledge further through subsequent means: staying up to date on researched, staying up to date on listserv, etc. But there have certainly been new things that came up not necessarily new in general, but new for me. I've been working more with research data management. So I have learned more about preparing data management plans, for example, which are a little more towards the beginning of the research workflow than where I normally come in. And then there have obviously been many developments at the policy level that I have learned about from the government initiatives, like the tri-agency open access policy on publications that really shaped how we work with faculty about making the research, open access, as well as the tri-agency research data management policy, which is still not completely released, but it's shaping a lot the research data management practices.

Madelin Burt-D'Agnillo 04:25
That's great. Awesome. And I guess just as a follow up question, where are you learning about research data management?

04:33

So, research, data management is not specifically part of my portfolio at work, but [there is a working group] that creates curriculum for staff development courses to improve staff and librarians' understanding of research data management, and specific practices. So that is how I intersect. And that's how I help myself learn more about it.

Madelin Burt-D'Agnillo 05:12
Awesome. Okay, right on. Now I'm gonna put question six in the chat, and then I'm going to share my screen. And usually I do it in this order. So that it's a little bit quicker. Alright, so this is a multi part question. So sort of the first part of this question is just generally speaking, are there any barriers that prevent you from adopting open science practices in your work, and given sort of your unique position you can speak specifically about in your role, or maybe you could speak more generally about some things that you think could impact somebody in your role. The next set of questions, we'll talk about other researchers, but you know, you can kind of speak about your experience here, or like a potential experience you could come up against. And then we're gonna look at this list. This list was developed from a similar study that was conducted in Australia. And so we can look at this list together and sort of determine if any of these barriers resonate with you or your work. And then finally, if there are any barriers that you identify, we could talk about how you might overcome them. So just to start back at the beginning of the question, just off the top of your head without even necessarily reading this list, are there any barriers that prevent you from adopting open science practices in your work?

06:48
So my work is to help make research open access? So in my case, I think question six and question seven are pretty similar in the sense that the barriers I see are the barriers that others may be experiencing. But I'm trying, I'll try not to look at this list so that it doesn't kind of skew me. So one of the things... I think the biggest barrier that I definitely encounter every single freaking day, is helping faculty or helping researchers in general, navigate copyright policies. And the fact that many journals still require full corporate transfer, when someone publishes, that's taken away the right and the ability of the author to then share their work openly, even if the author so chooses, is a really big barrier that many authors do not realize they are experiencing. So many sign away their copyright very easily. And obviously, they do not often have a choice, because once the manuscript is accepted for publication, they're presented with this document, and you're sign it or you don't get published. And really, we only started having these conversations, thanks to the tri-agency with access policy and publications that passed in 2015. So that's when authors realize that "oh I am under the funders requirements to make this article openly accessible, but I cannot because I signed my rights away. Now the publisher wouldn't let me or the publisher would let me if I pay article processing charges, which are upwards of $3,000." And which is I see the number four number one on your list, also lack of funding for open access publishing. And then where do I get that money? And of course, publishers have been the ones benefiting immensely from these policies, because they can then jack up those rates and essentially extort Open Access accessibility from authors. So this is something I see in my [role], because I try to get faculty to submit their research, we try to upload on their behalf, but we constantly run into these issues. And another related barrier is just the difficulty of understanding and navigating the copyright and permission landscape. Maybe, the permission was better, maybe it's not actually barrier, but figuring it out and reading that fine print and understanding and if it's not something that you do often, if you maybe publish once a year, you don't kind of have the I trained to read in that legalese. It can be very difficult and despite the support that we provide, despite our trained assistance, it's really difficult to even get faculty to consider investing even the minimum amount of time into this. So that is what I see in my role. Very, very commonly. Now, I think I'm ready to take a look at the list. So we talked about number one. So funding for open access publishing. Yes, it is a barrier. And I've talked to faculty who bring it up. And unfortunately, that may drive faculty members to publishing in a journal that is not

open access, if author processing charges are kind of in the way. Lack of credit in my institution to engage in open science... I don't know much about that lack of recognition in my field about developments... well. This is a little harder for me to address these because I do not work quite as closely in a specific field or with specific faculty members. Lack of mandates from funders, institutions or other regulators. Now, I'm not sure about that we do have mandates. The mandates we have in Canada are pretty toothless. So it's not so much lack of mandates is the lack of their enforcement probably. Lack of information about open science practices. This one definitely yes. I have encountered this a lot when talking with faculty and students about Creative Commons licensing. For example, when students submit their thesis, they're presented with an option to select a Creative Commons license with a thesis, it's an optional step, they can skip it. And they often do because they don't know what it means. But often what happens through my conversations with students is they will go to their advisor and ask "which license choice should I select?", and the advisor would be like, "I have no idea" and they would go talk to the classmates or to other people in department and no one knows what it is. So they walk away from this thinking that is just not something that is done in their discipline or in their field. And it's not something that they shouldn't do either. So I'm very grateful when they persist, and they reach out and we can have a conversation about the licenses and help them select one. Because then it means that they can further spread this knowledge. But if it's not something that your department does, it's not something that your lab or your supervisor does, then you're not likely to adopt these practices. Totally, um, lack of professional staff that provides support for open access practices. Perhaps a little hard for me to judge, I do know that in some departments where there is a dedicated admin staff that can help, for example, collect all have the accepted manuscripts from a team and send them for deposit into the repository, particularly if it's a big team that publishes a lot. And it definitely makes a difference. And we do work with a few teams like that in engineering, who publish a lot. And they have one single staff member who keeps track of everything and sends us everything and they are just so on point with meeting the requirement of their NSERC grants. Lack of research funding to support them. So that is kind of related to number one, probably, lack of training required to implement open open source practices. Yes, there's definitely a bit of that, particularly as more and more requirements are coming out, for example, preparing your data management plan and things like that. All of that requires some training all that requires setting aside some, some headspace and some brain power to introduce those. Lack of supporting infrastructure. This will be very specific to the country or the institution you're talking about. For example, at U of T, we're lucky to have a lot of repositories and a lot of staff that support them. We have dataverse, we have TSpace, we have dedicated Scholarly Communications librarians and Research Data Management librarians. So in that sense, we definitely have infrastructure. Now smaller institutions might not have that, smaller institutions might not have consortiums that they can rely on definitely be the case there. Lack of time to engage in lack of time to learn. Yes, this is related to that headspace that I mentioned before. And also related, I think, to the perceived culture in the department and in the particular lab. Lack of expertise engaging up and says practices like assignment of metadata. And this is related to to learning and to time to engage. So breaking the mould, breaking habits doing things in certain way definitely always require some extra time and extra expertise. Even when support is available, for example, through libraries, like in our case, it does take extra effort on researchers behalf to reach out and take advantage of that. Researchers are discouraged from engaging in open science practices by their colleagues. I have not necessarily witnessed that specifically, it's less about them being discouraged and more about maybe being not encouraged. That the same thing. Students are discouraged from engaging in science practice, buy their thesis supervisors, once again, I do not have any personal experience or anecdotal evidence of that. But definitely, students and early career researchers are definitely at a disadvantage in terms of selecting Open Access journals. Even if they would like to publish in them, because they are usually under pressure to build a portfolio for tenure and promotion. And depending on the culture in the department, if the department expects you to publish in specific journals, of which maybe there are very few in the field, and if those journals don't happen to be open access, then that is a really difficult choice to make. Do you follow what's best for your career? Do you follow what's best for your kind of warm and fuzzy feeling of making your research accessible. And, of course, early career researchers cannot be faulted for making those choices. And in my experience, in working with faculty and researchers who choose to make the research open access, those are usually not early career researchers, usually they are past the tenure and promotion review, where they are free to select whatever journal they want, and they can do whatever they want. So their hands are a little bit more untied. So open science community is intimidating. Wow, I have not heard that one. Maybe I'm too too deep in the open science community. I think maybe there is a bit of a bit of truth to it in the sense that there is a lot of technology that is associated with open science. When we start talking about open science, we start talking about DOIs and ORCIDs and repository use and data transfer and metadata. And I think that a related talk may be intimidating and overwhelming. I don't know that it's the openness itself necessarily. I think it's just us librarians getting too excited about the technology. Researchers don't want to be told how to do their research. Of course, yes, I have seen a lot of pushback in presentations that I have given. That has to do with mandates. So whenever any sort of mandate or any sort of commitment, there tends to be pushed back, even if researchers themselves would voluntarily commit to certain things, they just don't like to be committed to those things. Yeah, there's definitely a bit of that. Lack of interest from researchers? Yes, perhaps lack of interest or lack of time. And once again, that is also very discipline specific. I do not perceive any barriers. That is not true. I've talked to a lot of barriers. Yeah. Is there anything else you would like me to add or any other points that you would like me to dwell on?

Madelin Burt-D'Agnillo 18:36
I mean, that was awesome. I think you did a really good job there positioning yourself and saying what you're familiar with, without over speaking, speaking on behalf of someone else. But the sort of follow up question that I think you've done a great job of answering already is, are there barriers that prevent researchers, including students, faculty, other librarians or administrators in your discipline from adopting open science practices? So I think you've answered that perfectly well, and given a lot of examples, if there's anybody on that list that you think you have more to speak about, go for it. But I think we could just move to question seven B, or six B is the same. What could help you or others overcome some of the barriers presented on this list in your experience?

19:30
Yeah, I agree. I think we talked a lot about the barrier. So let's talk about opportunities. So what can be done? I think in all of these, in many, if not most of these cases, there's just an overall kind of lack of motivation. Since the inception of open access, which was, what 20 years ago, it was always the idea that you're doing it for the sake of being a better person and contributing to the advancement of research for the sake of that warm and fuzzy feeling. And unfortunately, that hasn't changed much. And that's often just not enough, not enough for researchers to carve out the time in their busy schedules to change their ways, to put together a data management plan or to collect their accepted manuscripts and put them in the repository. And they cannot be faulted for that they're super busy and super stretched. So the question then is, how can we add, in addition to this internal warm and fuzzy motivation? How can we add some external motivation to promote that, so far as motivation will have seen has come in the form of mandates, and those mandates come attached to funding. So if you get public funding, you're expected the results of your research to be made publicly available, that's perfectly fine and understandable. And there is a great logic behind it, and I'm behind those mandates as well. But then there is also lack of enforcement of those mandates, so they're not actually making a difference, making as much difference as they could. And even if there was more enforcement, is this really kind of the positive motivation that we want to see? So I would like to see more motivation from

within the institution itself. In tenure and promotion committees, more emphasis placed on open access, publishing, and not just on publishing in high impact journals, etc. As well as through the annual reviews, more emphasis on not just sharing a CV with a bibliography of your research, but actually your research, I make a point of putting as much as possible of my research into our institutional repository so that when I prepare my annual review, everything that I said is linked. This is the article I published, here's the link. This presentation I gave, here's the link. [...] And I find that even for me, this takes a little bit of extra effort if I slip, and if I don't do it, I only realize that at the end of the year, when I'm preparing this report, I'm like, "Oh, yes, I could have done this a year ago". And I can only imagine how much further down the priority list. It is for every one that was not the [role]. So if this was more of an expectation, or if this expectation was made more clear, as part of the annual activity report, then this may help. And I don't know much about how those reports work from faculty, but imagine that something can be done there as well.

Madelin Burt-D'Agnillo 22:45
Brilliant. Okay. I really liked that answer. Thank you again. So I'm going to stop sharing my screen. And we'll move into question eight, which is one that, again, feels a little bit misplaced in this interview, but it's on the list. So I'll ask it anyway. Have you received research support relating to open science practices from the University of Toronto Library System? And if so, at what stage in the research?

23:14
Yeah, that's a little weird. I feel like I'm my own my own support system. But in a way, in a way, yes. [...] Yes, now, I know a lot about how to put together a research data management plan. And I will certainly be doing that as part of my own research. And one of the user groups we were addressing when we were putting together that curriculum is librarians and staff who work on their own research. So if we made it even a little bit of a dent there, then I consider myself lucky and grateful. And of course, just for working in the system, and from knowing the supports and how to select a journal and how to read through journals, copyright and self-archiving policies, all of those things will make it much easier for me when time comes to share the results of my research.

Madelin Burt-D'Agnillo 24:44
Awesome. Great. Okay, and we're almost at the end. This question is nine of 10. In your opinion, how important are open science initiatives at the University of Toronto?

24:58
Well, that's a very big question. Because the University of Toronto is very big. And also, how do you define being important, right? I think the only way to look at this question was to look at it in more kind of narrow scope of individual departments or individual programs. I know that there are some fantastic initiatives at U of T, like the Structural Genomics Consortium that absolutely made it their mandate to share their research as early and as openly as possible. So they share not just the final results, and not just the data, but also their lab notebooks, they essentially share the research as it is happening. And it has made such an impact and such a difference all over the world for others who work in the same space and who know that, "oh, this research has been conducted, I don't need to do the same thing, because I know that I would be duplicating someone's efforts, how about instead, I collaborate with them, so we can do together faster." So this is a really great example. But that is a really small initiative. That is pretty much driven by a single person with a lot of vision and dedication to open science. I have seen institutions pass an open access mandate for faculty. And I have not yet seen, or at least not in Canada, an actual impactful one. Because all of them end up being rather smooth around the edges and end up being more recommendations really, than mandates. At U of T, we don't have a university-wide one. But for example, OISE has one that I'm sure not many people are aware of anymore, it was

kind of hot back in the day [...]. And I think most people just forgot about about it. And that is what happens, it's really hard to do anything at the university scope, considering the size and the hierarchy at U of T so I think the most we can hope for is for individual departments to drive to forge the way and and for others to follow.